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BRIEF REPORT

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Teaching children with autism spectrum disorder to mand for answers to questions via telehealth: A caregiver implementation

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Abstract

Manding for information is important because information is a valuable part of learning and interacting with our environment, allowing for more autonomy (i.e., access reinforcement). Despite the importance of this skill, individuals with autism spectrum disorder (ASD) often do not engage in manding for information without explicit instruction. In the current study, we coached caregivers via telehealth to teach three children with ASD to mand for information using the phrase "I don't know, please tell me" (IDKPTM). All three participants independently engaged in the IDKPTM response for information during intervention and post intervention generalization probes. The results of this study extend the findings of previous research by demonstrating caregivers coached via telehealth can teach their children to mand for more information using the "IDKPTM" phrase. Additionally, the results of this study reflect the benefits of involving caregivers to implement evidence-based interventions while being coached through telehealth.

KEYWORDS
autism, caregiver coaching, mands for information, telehealth

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1 | INTRODUCTION

In the last few years, especially since the beginning of the COVID-19 pandemic, there has been an increase in services, including behavioral intervention services, provided via telehealth (United States Department of Health and Human Services (HHS), 2021). One arrangement of telehealth services that has been employed by behavioral health providers is coaching caregivers to implement interventions based on the principles of applied behavior analysis (ABA). Caregiver-implemented interventions have been shown to be effective and may be more efficient with the caregivers serving as a natural change agent in their child's daily life. Equipping caregivers with tools and knowledge based on evidence-based practices to teach their child vital skills can create lasting and positive changes in families' lives (Jeong et al., 2021).

Telehealth provides a unique opportunity to engage with caregivers without physically imposing on the family. Within the field of ABA, interventions implemented by a caregiver range from teaching daily living skills to treating problem behavior with children diagnosed with autism spectrum disorder (ASD) or other neurodevelopmental disorders (Gerow et al., 2021; Lindgren et al., 2016). The variety of skills that can be taught through caregiver coaching via telehealth may allow for clinicians and researchers to reach more families, including those in rural communities.

One skill that may be addressed using this caregiver coaching model is manding (i.e., requesting) for information. Throughout our daily lives, we use information to navigate our environment and engage with other individuals more effectively. We learn to use our verbal behavior to access information that we did not previously have by asking other people for this information. For example, if a driver is unable to see if they are clear to cross an intersection, they may request this information from the passenger. We mand for information and respond to mands for information frequently. This skill not only allows for more effective navigation of one's environment, but also additional autonomy. A child who can request the location of the scooter rather than request for their caregiver to give them the scooter, can access the scooter ride quicker since they do not need to wait for their caregiver to obtain it. Despite the importance of this skill, individuals with ASD often do not engage in manding for information without explicit instruction.

Across the literature, there are several interventions that target teaching individuals with ASD to mand for information including echoic prompting (e.g., "say what is it"), naturalistic teaching, and discrete trial training. The topography of the mands for information also differ across studies, such as teaching a child to ask a question such as "What is it?" (Lechago et al., 2013) or "Where's "*toy*"? (Somers et al., 2014). In some instances, we are asked questions that we do not know the answer to, and we may need to request this information. This often occurs when an instructor is evaluating our knowledge on a certain topic (e.g., "What is the capital of California?"). Ingvarsson and Hollobaugh (2010) investigated teaching these types of requests to four boys diagnosed with ASD. Specifically, researchers taught children using echoic prompting and constant prompt delay to ask for additional information by using the phase "I don't know, please tell me" (IDKPTM). This study was conducted in a school setting and included a variety of questions related to personal information, general knowledge (e.g., "Where do you buy clothes?") and academic skills (Ingvarsson & Hollobaugh, 2010). The findings of this study indicate that mands for information can be taught using echoic prompting across a variety of questions.

Within the current literature base on manding for information, there are few studies that incorporate caregivers as primary interventionists. Therefore, the purpose of the current study was to replicate and extend the findings of Ingvarsson and Hollobaugh (2010) by specifically having a caregiver implement the intervention while receiving coaching from the researcher over telehealth.

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2 | METHOD

2.1 | Participants

We recruited three children with ASD and their caregivers to participate in this study. The participants included two males, Alex (8 years) and Aden (7 years), and one female, Ida (9 years). Alex and his caregiver were Hispanic, Aden and his caregiver were Black, and Ida and her caregiver were both White and Hispanic. Participants were recruited from a larger research project on telehealth coaching with caregivers of children with ASD. All participants had an established echoic, manding and tacting repertoire, and difficulties or deficits in intraverbal skills. These skills were evaluated during a pre-test to assess for inclusion prior to beginning the study.

3 | SETTING AND MATERIALS

Caregivers implemented sessions in their homes, either in the kitchen (Alex) or the child's bedroom (Aden and Ida). Researchers provided coaching via telehealth from a university campus or in their own respective homes. Materials for this study included a secure video conferencing platform, technology with video and internet capabilities (e.g., laptop, tablet) for both participants and researcher, data sheets with randomized trial order, and preferred items selected by the child before sessions.

3.1 | Measurement

We recorded data on the number of trials with independent engagement in the IDKPTM response and independent correct answers to questions. A response was recorded as independent if the IDKPTM response was initiated within 5 s of the question being asked by the caregiver. A response was recorded as prompted if the caregiver provided the vocal prompt "say IDKPTM".

To assess for interobserver agreement (IOA) the primary data recorder collected data live during telehealth sessions for all participants. A second researcher simultaneously, but independently, collected data for least 50% of sessions during baseline, intervention and generalization probes for each child. A trial was scored as an agreement if both researchers marked the same symbol (i.e., "+" or "−") for the IDKPTM response and answer to the question. The percentage of trials with agreements for each session was calculated.

We collected caregiver implementation fidelity data during a minimum of 50% of sessions for each child during baseline, intervention, and generalization probes. The components of implementation fidelity for the caregiver included the caregiver providing the instruction, following the steps corresponding to each condition, such as providing prompts at the correct time delay (i.e., 0 s or 5 s), and providing access to a 1 min break after the child responded to five questions. We did not provide explicit caregiver training for this specific protocol; however, caregivers had already received previous training on implementing discrete trial training with their child. In addition to caregiver fidelity, we also recorded the researcher implementation fidelity during at least 50% of sessions for each participant. The components of implementation fidelity for the researcher included the researcher providing the instruction, collecting data, and providing feedback to the caregiver during each condition.

At the completion of the study, we sent an 18-item social validity questionnaire to each caregiver. The survey contained 16 Likert-scale questions and two open ended questions based on the Treatment Acceptability Rating Form-Revised) by Reimers et al. (1991). The results of IOA, treatment fidelity and social validity can be found in Table 1.

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TABLE 1 Results of treatment fidelity, interobserver agreement and social validity averages.

	Section			
Participant	Caregiver treatment fidelity	Researcher treatment fidelity	Interobserver agreement	Social validity
Alex	96.33% (89%–100%)[a]	100%	98.25% (80%–100%)	4.15/6
	Across 100% of sessions	Across 50% of sessions	Across 75% of sessions	
Aden	100%	100%	99.33% (80%–100%)	6/6
	Across 100% of sessions	Across 84% of sessions	Across 58.35% of sessions	
Ida	100%	100%	98.5% (80%–100%)	4.15/6
	Across 100% of sessions	Across 88% of sessions	Across 78.5% sessions	

[a]average across all phases.

3.2 | Experimental design

A nonconcurrent multiple baseline design across participants was used to evaluate the effectiveness of IDKPTM training to increase mands for information and correct responses.

3.3 | Procedures

Researchers facilitated 30–45 min sessions, one to two days per week via telehealth. Caregivers implemented the intervention during all sessions. Caregivers provided instruction directly to their child and received coaching from the researcher. During each session, the researcher reviewed the instructions with the caregiver and displayed five questions at a time on the screen through the screen sharing function. Instructions were given to caregivers in a combination of written and verbal formats. The caregiver was instructed to make sure the child was unable to see the screen when the researcher was screen sharing. Each question was considered a trial and five trials were conducted per session.

3.3.1 | Pretest and question selection

The questions included in this study were determined by a pretest of 40 questions that targeted general knowledge (e.g., “Where do you buy clothes?”), personal information (e.g., “When do you go to school?”) and academic questions (e.g., “How much is a quarter?”). Questions answered incorrectly at any point during the pretest were included. The caregiver provided descriptive praise for correct answers and neutral statements for incorrect or no response to the question such as “good try” or “that's okay.” The caregiver selected 20 questions from the unknown questions. These questions were then randomly assigned to Unknown Set 1 and Unknown Set 2.

3.3.2 | Baseline

We instructed the caregiver to ask questions from Unknown Set 1. Each session consisted of five previously randomized questions. Three sessions were completed each day for a total of 15 trials. Following the completion of each 5-trial session, the child received a 1 min break. Following a correct answer to the question, the caregiver provided descriptive praise. If the child emitted the response “I don't know” or “I don't know please tell me”, the

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caregiver provided praise and the answer to the question. If the child gave an incorrect answer to the question or did not respond within 5 s, the caregiver asked the next question. Unknown Set two questions were also probed during baseline for each participant with the same procedures as baseline.

3.3.3 | Intervention

This phase was similar to baseline, except the caregiver provided a vocal prompt to teach the child to engage in the IDKPTM response. For the first three sessions, the caregiver provided the prompt on a 0 s delay. For the remaining sessions, the caregiver provided the prompt after a 5 s delay. If the child engaged in the IDKPTM response independently or after the prompt, the caregiver provided praise and the correct answer.

For all three participants, after multiple sessions, we reduced the number of targets from 10 questions in Unknown Set 1 to five questions. This decision was made because the participants had mastered the IDKPTM response, which was the primary aim of the study, and were failing to finish the study based on mastery of the questions (e.g., "How much is a quarter?"). We used a computer-generated randomizer to select the five remaining questions. The session procedures remained the same, the only alteration was that the same five questions were asked every session, whereas before they were randomized from a pool of 10 questions.

3.3.4 | Generalization probes

Generalization probes included questions from Unknown Set 2 and followed baseline procedures. The caregiver praised correct answers, provided the correct answer contingent on the IDKPTM response, and refrained from providing any prompts.

4 | RESULTS

Results for all three participants are displayed in Figure 1. Alex's results are displayed in the top panel of Figure 1. During baseline, Alex had zero correct answers to the questions, zero IDKPTM responses for target (Unknown Set 1) and generalization (Unknown Set 2) questions. After IDKPTM teaching trials, Alex began to independently engage in the IDKPTM response with the target set (range: 20%–80% independent IDKPTM responses), shortly followed by independent engagement in the IDKPTM response during generalization sessions (range: 0%–80%). As Alex began to respond correctly to the questions, he engaged in fewer instances of the IDKPTM response. Although the primary purpose of the study was to teach the children the IDKPTM response, we were also interested in the participants' mastery of answering the questions. We wanted to ensure the IDKPTM response came under appropriate stimulus control of not knowing the answer. Therefore, starting in session 40 (denoted by the phase change line), we reduced target questions from 10 to five targets. After reducing the number of target questions, Alex's correct answers to questions began to increase, meeting mastery criterion by session 60 with 87% (range: 80%–100% correct responses) accuracy across three sessions.

Aden's results are displayed in the middle panel of Figure 1. During baseline, he did not engage in any IDKPTM responses or correct responses to questions. After IDKPTM teaching trials were completed, Aden independently engaged in the IDKPTM response with 80%–100% accuracy until session 21. During generalization sessions, Aden engaged in the IDKPTM response independently and accurately to the unknown questions (range: 20%–60%). Due to a break in sessions at the caregiver's request, we did not meet for 6 weeks, denoted by an asterisk during intervention. After resuming sessions, we decreased the number of targets to five at session number 34, denoted by

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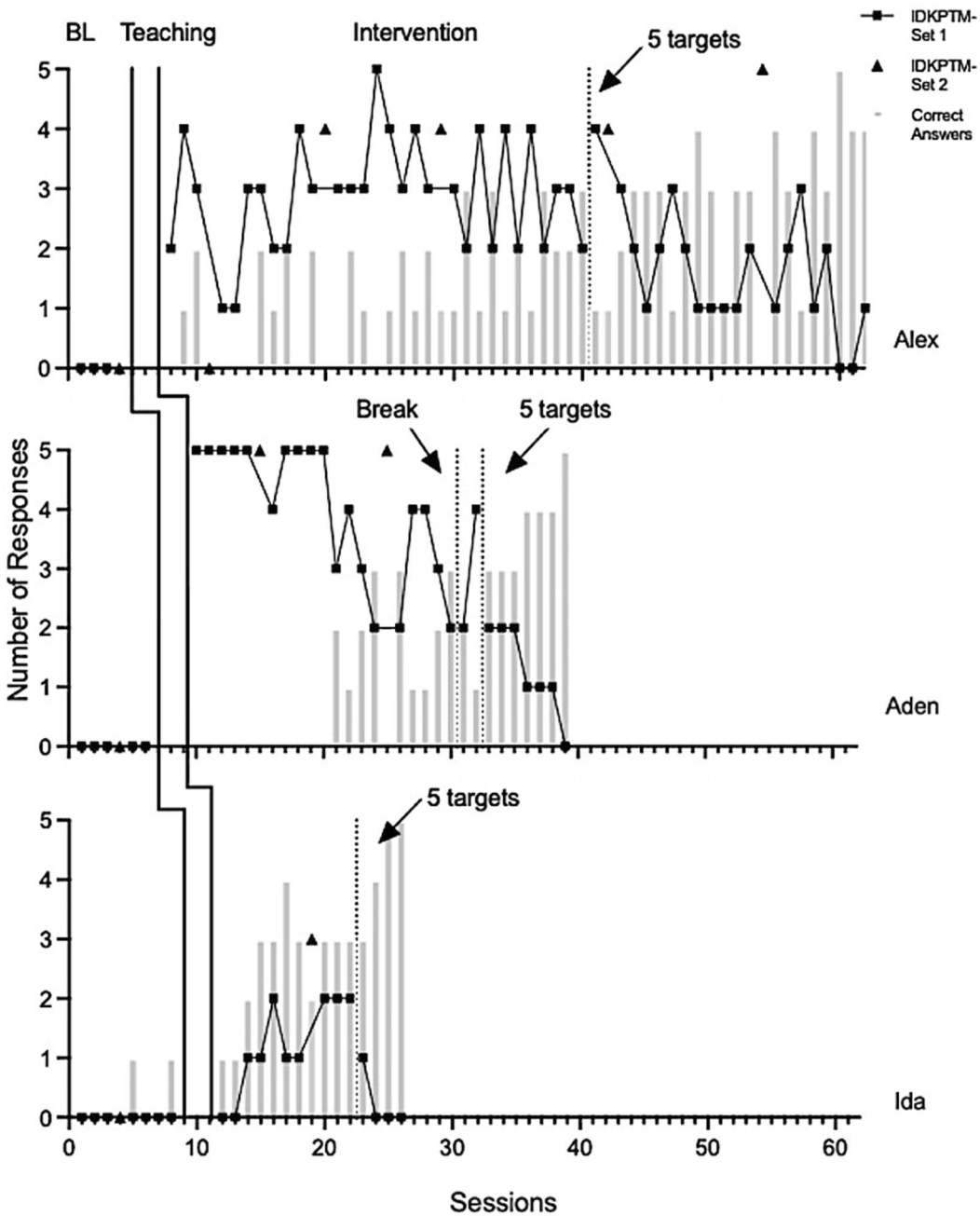

FIGURE 1 Alex, Aden, and Ida's multiple baseline graph of IDKPTM responses, correct answers and generalization probes. IDKPTM Set 1 are questions selected for intervention and IDKPTM Set 2 are questions used in generalization probes. The phase change lines between baseline and intervention are sessions implemented with a 0 s time delay. The asterisk marks the session where Aden resumed sessions following a break. The data paths reflect an inverse relationship of the IDKPTM response and the acquisition of correct answers to questions. IDKPTM, "I don't know, please tell me".

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the phase change line. By session 39, Aden had reached mastery criterion of correct responses (range: 80%–100%), while independently engaging in the IDKPTM response for one of the five targets.

Ida's results are displayed in the bottom panel of Figure 1. During baseline, she did not engage in the IDKPTM response; however, she did have two instances of correct responses to the target questions in Unknown Set 1 (range: 0%–20% correct responses) and did not engage in any correct responses during the generalization session. Following IDKPTM teaching trials, Ida emitted some independent correct responses to target questions and after session 14, she began to emit the IDKPTM response independently (range: 0%–20%). As intervention continued, she answered more questions correctly. During the generalization session in intervention, she correctly answered two questions (n = 40%) and used the IDKPTM response independently for the other three trials (n = 60%). We reduced the number of target questions to five before session 24 denoted by the phase change line. By session 26, Ida reached mastery criterion (range: 80%–100% correct responses).

5 | DISCUSSION

Our study provides a replication and extension, with a telehealth caregiver implementation, to the previous findings of Ingvarsson and Hollobaugh (2010). Inclusion of caregivers is important because caregivers are provided with effective strategies to use with their child that are appropriate in different situations. In this study, the caregiver learned to effectively use vocal prompts and a prompt delay to promote independence which can be used to teach a variety of important skills to children. Furthermore, telehealth can be an efficient and less intrusive method of service delivery that can include meaningful involvement of the caregiver. Telehealth services allow for more accessibility and flexibility to best meet the needs of each family. This was especially notable during the COVID-19 pandemic as telehealth was often the only way for families to access services.

By teaching children with ASD to mand for information, there is an opportunity for a child to expand their vocabulary and verbal repertoire (Betz et al., 2010). This expansion could lead to more efficient and effective navigation of different environments, while also providing opportunities for increased social interactions (Lechago et al., 2013). In this study, we observed that across all children, independent mands for information increased during intervention and generalization sessions. Specifically, all children independently engaged in the IDKPTM response with over 85% accuracy (range: 60%–100%) during the generalization probes when the answer was not known (i.e., when the child did not emit the correct response to the question). We also observed with the secondary aim of the study, the acquisition of the correct response to questions asked in Unknown Set 1, participants had mastered engaging in the IDKPTM response before reaching mastery criterion for the correct answer. By reducing the number of trials, we observed all participants reach mastery criterion of providing the correct answers to the questions in Unknown Set 1 in fewer sessions, compared to the number of sessions conducted with 10 questions during intervention. Further, the intervention procedures were reported to be socially valid and positively viewed by all three caregivers. Two caregivers noted on the questionnaire anecdotal examples of their children manding for information in different situations and settings. These results reflect the benefits of providing services via telehealth and having caregivers serve as the primary implementer.

There are some limitations of this study worth noting. First, although we intended to conduct sessions for all participants concurrently, there were time gaps between sessions for each participant due to family illnesses and changes in familial dynamics (i.e., new baby). Each of these gaps occurred after we initiated the intervention phase for all three participants, limiting the effect of these gaps on the design of the study. Second, we acknowledge that the IDKPTM response may not be the most relevant response in all circumstances. Despite this, this phrase is acceptable across a variety of situations, and is likely to contact reinforcement in both casual and formal conversational situations and for different types of questions. Researchers should continue to investigate phrases that are both functional and socially appropriate across contexts and should evaluate methods to teach children

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how to choose the most appropriate question for each context. Third, we did not use a control set of known questions. A control set would have would allowed for a clear differentiation in the participant's use of the IDKPTM response to unknown questions versus known questions. Although we did not include a known set, the participants decrease in the use of the IDKPTM response as they mastered unknown questions demonstrates that the IDKPTM response came under appropriate stimulus control of not knowing the answer. Finally, in this study we did not assess the generalization of manding for information across settings and individuals. Although we did observe generalization of the IDKPTM response to unknown questions asked by the caregiver, to further test and support generalization of the skill, future research should evaluate across settings and individuals.

In conclusion, the results of this study showed that coaching via telehealth to teach caregivers to implement mand-training was effective in increasing each child's independent mands for information. This study further contributes to the area of research related to mands for information and children with ASD. Additionally, the results of this study reflect the benefits of involving caregivers to implement evidence-based interventions while being coached through telehealth.

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CONFLICT OF INTEREST STATEMENT

There is no conflict of interest.

DATA AVAILABILITY STATEMENT

Due to the nature of the research, due to restrictions, supporting data is not available.

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